Dilations and Similar Figures in Practice

Scale drawings, models, shadows and enlargements — every one built from a single scale factor

- Dilation about the origin by factor k: $(x, y) \to (kx,\ ky)$. About a centre (h, j): $(x, y) \to \big(h + k(x - h),\ \ j + k(y - j)\big)$.
- A dilation multiplies every length by k and leaves every angle alone, so the image is similar to the original.
- Units are stated in every problem — give your answer in the unit asked for.

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1. On a design grid marked in centimetres, point $A(6, -4)$ is dilated about the origin by a scale factor of 0.5. Write the coordinates of A'.

A' (________, ________)

2. A photograph 12 cm wide is enlarged so that the copy is 30 cm wide. Find the scale factor of the enlargement.

Answer: ________________

3. A blueprint is drawn at a scale of 1 cm to 40 cm. A wall measures 8.5 cm on the blueprint. Find the real length of that wall.

Answer: ________________ cm

4. On a plan marked in metres, a triangle has vertices $P(2,1)$ and $Q(6,4)$. The plan is dilated about the origin by a scale factor of 3, and Q maps to $Q'(18,12)$.

(a) Write the coordinates of P'. P' (________, ________)

(b) Find the length PQ. ________

(c) Find the length $P'Q'$. ________

5. A model car is 18 cm long. The real car it copies is 4.5 m long, which is 450 cm. Find how many times longer the real car is than the model.

Answer: ________________

6. A student 1.6 m tall casts a shadow 2.4 m long. At that same moment a flagpole casts a shadow 13.5 m long. The student and the flagpole form similar right triangles with their shadows. Find the height of the flagpole.

Answer: ________________ m

7. A logo on a screen is dilated about the point $C(2, 3)$ by a scale factor of 4. The point $D(5, 1)$ maps to D'. Write the coordinates of D'.

D' (________, ________)

8. An architect scales a courtyard plan by a factor of 2.5. The original courtyard is a rectangle 12 m by 8 m.

(a) Find the perimeter of the scaled courtyard.

Answer: ________________ m

(b) Find the area of the scaled courtyard.

Answer: ________________ m^2

(c) The perimeter grew by a factor of 2.5 but the area did not. Explain why the same scale factor acts differently on the two measurements.